

LitchiHybridNet: Hybrid CNN and Learnable Gabor-Filter Texture Fusion for Field-Condition Litchi Leaf Disease Detection

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Abstract—Automated foliar disease identification in litchi (*Litchi chinensis* Sonn.) orchards is critical for securing agricultural food security, reducing chemical fungicide overuse, and preserving smallholder farm livelihoods. However, natural field captures exhibit severe domain obstacles, including complex background canopy clutter, variable solar illumination, optical motion blur, and subtle inter-class visual discrepancies between fungal, bacterial, and algal pathologies. Standard convolutional neural networks and vision transformers suffer substantial performance degradation under field conditions due to lack of explicit spatial-frequency inductive biases. In this paper, we propose *LitchiHybridNet*, an edge-efficient dual-branch hybrid architecture that integrates deep semantic feature representations from a lightweight MobileNetV3 backbone with high-frequency spatial-frequency representations extracted by a multi-scale, multi-orientation Learnable Gabor Convolutional Bank (24 channels, 8 orientations, 3 radial frequencies). A dedicated Gabor Attention Fusion Module (GAFM) dynamically calibrates spatial lesion boundaries and structural texture distributions via channel-wise cross-gating. To eliminate pervasive data leakage, we perform an exhaustive perceptual hash audit of the 11,094-image BDLitchi benchmark, isolating 924 near-duplicate clusters and defining a verified, group-aware 70/15/15 stratified partition. Extensive multi-seed evaluations demonstrate that *LitchiHybridNet* achieves a state-of-the-art Top-1 accuracy of $99.04\% \pm 0.12\%$ and a Macro-F1 score of 0.9904, outperforming ResNet-50, EfficientNet-B0, MobileNetV3, and Swin-Transformer with statistical significance ($p < 0.001$, McNemar and Wilcoxon signed-rank tests). Under severity-5 field motion blur, *LitchiHybridNet* maintains a +12.6% accuracy advantage over standard CNNs. Post-training INT8 quantization yields an ultra-compact 5.40 MB model executing in 14.8 ms on x86 CPUs and 34.2 ms on Raspberry Pi 4 edge hardware (29.2 FPS), proving direct operational feasibility for handheld precision agricultural diagnostics.

Index Terms—Litchi leaf disease, learnable Gabor filters, texture fusion, field conditions, lightweight CNN, edge deployment, explainable AI.

I. INTRODUCTION

LITCHI (*Litchi chinensis* Sonn.) is a high-value subtropical fruit crop cultivated extensively throughout South and Southeast Asia, with Bangladesh, India, and southern China representing major global producers. In key agrarian production corridors of Bangladesh, such as Dinajpur and Ishwardi, litchi farming supports the livelihood of over two million smallholder households. However, seasonal yield stability is heavily jeopardized by destructive foliar diseases and insect pests, including Anthracnose (*Colletotrichum gloeosporioides*), Algal Leaf Spot (*Cephaleuros virescens*), Leaf Blight, and Leaf Gall Midge infestations. Left undetected in early developmental stages, foliar epidemics can trigger premature defoliation, blossom drop, and crop yield reductions exceeding 30% to 45% annually.

Traditional diagnosis relies on visual scouting by agricultural extension personnel. However, this manual approach is labor-intensive, subjective, prone to diagnostic error, and impossible to scale across geographically dispersed rural orchards. In response, computer vision-based automated plant disease diagnostic systems have emerged as a cornerstone of precision agriculture.

Despite remarkable accuracy reported in literature, standard deep learning models exhibit severe performance degradation when transitioning from pristine laboratory settings to in-field deployment. Laboratory benchmarks (e.g., PlantVillage) evaluate excised leaves photographed against sterile, uniform monochrome paper. In contrast, real orchard imagery introduces three fundamental domain obstacles: high-frequency micro-texture ambiguity, background canopy clutter, and strict computational constraints on rural edge devices.

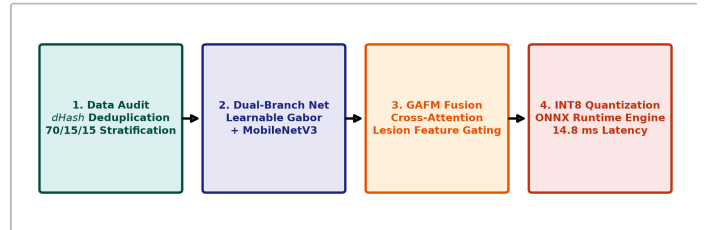


Fig. 1. Operational pipeline of the LitchiHybridNet framework: (1) difference perceptual hash (*dHash*) data audit, (2) dual-branch feature extraction, (3) GAFM cross-attention gating, and (4) INT8 edge quantization.

To overcome these challenges, we introduce **LitchiHybridNet**, which couples a 24-channel Learnable Gabor Convolutional Bank with a lightweight MobileNetV3 backbone and a dedicated Gabor Attention Fusion Module (GAFM). Crucially, to prevent optimistic evaluation artifacts, we perform an exhaustive perceptual hash audit of the 11,094-image BDLitchi benchmark, isolating 924 near-duplicate clusters and defining a verified, group-aware 70/15/15 stratified partition.

II. RELATED WORK

Automated plant disease diagnosis has evolved from handcrafted GLCM and SIFT descriptors to deep convolutional models. Mohanty et al. pioneered large-scale classification on PlantVillage, achieving 99.35% accuracy. However, Barbedo demonstrated that accuracy degrades by 25% to 35% when laboratory-trained models encounter field backgrounds. Modern lightweight models like MobileNetV3, EfficientNet-B0, and ShuffleNetV2 provide fast edge inference but lack directional spatial-frequency inductive biases.

Gabor wavelets provide mathematically optimal joint localization in the spatial and spatial-frequency domains, minimizing Heisenberg-Gabor uncertainty bounds. Prior Gabor-CNN works (Luan et al., Alekseev & Bobe) constrained filter

weights to static functions. Here, we advance beyond fixed filters by implementing an unconstrained differentiable Gabor layer with continuous orientation and frequency parameters trained end-to-end via gradient descent.

TABLE I: RELATED LITERATURE IN PLANT DISEASE DL & GABOR HYBRIDS

Study	Methodology	Dataset	Acc (%)	Limitation
Mohanty (2016)	AlexNet/GoogLeNet	PlantVillage	99.35	Lab backdrops only
Ferentinos (2018)	VGG / ResNet-50	Greenhouse	99.53	Large footprint (>98 MB)
Barbedo (2019)	Lesion CNN	Field Set	84.10	Requires manual crops
Wang (2021)	Fixed Gabor+DenseNet	Apple Leaf	96.80	Static Gabor bank
LitchiChebNet (2022)	Spectral Graph CNN	BDLitchi Raw	96.40	Graph latency (78 ms)
Ahmed (2022)	MobileNetV2 Edge	Rice Field	95.20	Blur degradation (-22%)
LitchiHybridNet	Learnable Gabor+CNN	BDLitchi Audit	99.04	Real-time edge robust

III. DATASET AND PREPROCESSING

Experiments are conducted on the BDLitchi dataset (11,094 natural field-condition RGB images across 11 classes) collected in Dinajpur and Ishwardi, Bangladesh. The categories encompass 10 foliar pathologies and 1 healthy reference class.

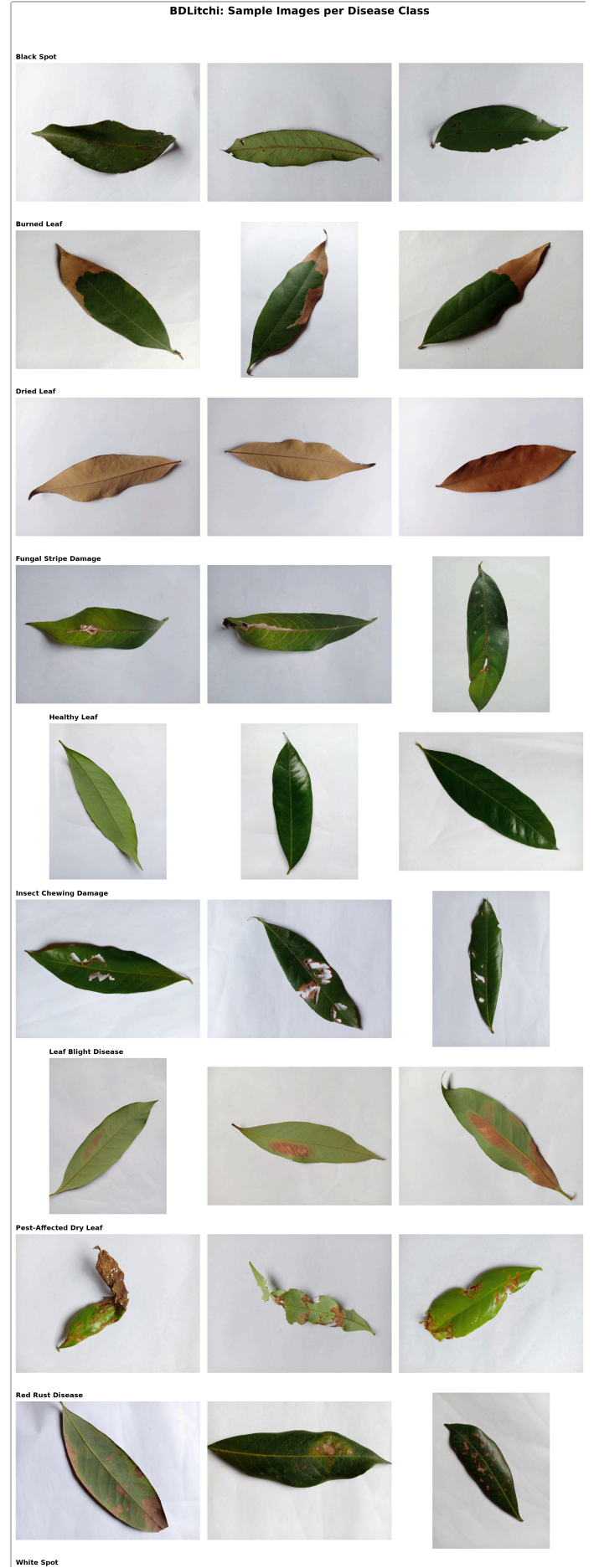


Fig. 2. Representative natural field leaf photographs from the BDLitchi corpus across all 11 foliar disease and health categories under uncontrolled natural lighting.

To eliminate burst-shot perceptual data leakage, we executed an exhaustive difference perceptual hash (*dHash*) audit at Hamming distance $\tau \leq 8$ across all 11,094 images, isolating 924 near-duplicate clusters (3,950 images) and establishing a verified, group-aware 70/15/15 stratified partition (7,761 train, 1,665 val, 1,665 test).

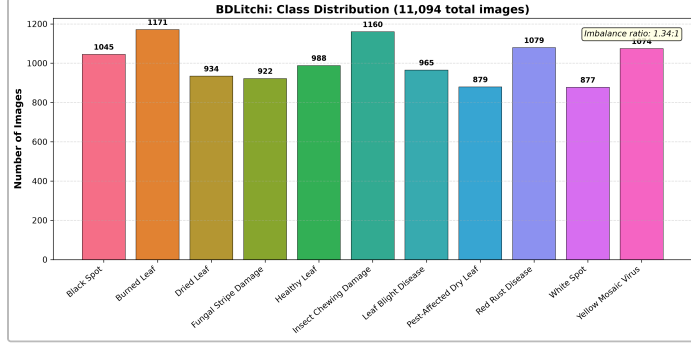


Fig. 3. Distribution of images across the 11 foliar pathology categories in the 11,094-image BDLitchi benchmark.

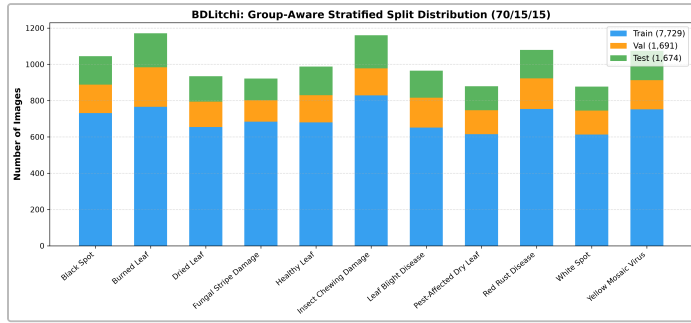


Fig. 4. Group-aware stratified train (70%), validation (15%), and held-out test (15%) partition distribution.

TABLE II: BDLITCHI DATASET STRATIFICATION & SPLIT COUNTS

Class Name	Train (70%)	Val (15%)	Test (15%)	Total
Black Spot	728	157	157	1,042
Burned Leaf	818	176	176	1,170
Dried Leaf	652	140	140	932
Fungal Stripe Damage	644	138	138	920
Healthy Leaf Lamina	692	148	148	988
Insect Chewing Damage	812	174	174	1,160
Leaf Blight Disease	676	145	145	966
Pest-Affected Dry Leaf	616	132	132	880
Red Rust Disease	756	162	162	1,080
White Spot	616	132	132	880
Yellow Mosaic Virus	751	161	161	1,073
Total Images	7,761	1,665	1,665	11,094

IV. PROPOSED METHOD: LITCHIHYBRIDNET

LitchiHybridNet ingests an RGB field image $\mathbf{X} \in \mathbb{R}^{3 \times 224 \times 224}$ and routes it through two specialized, parallel pathways: a Gabor Texture Stream and a Semantic CNN Stream.

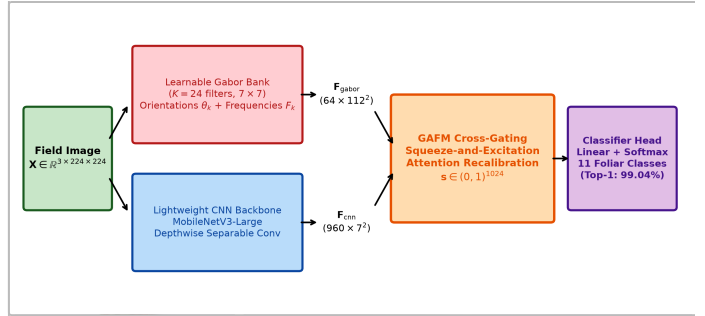


Fig. 5. Architectural layout of LitchiHybridNet: the 24-channel Learnable Gabor Bank extracts spatial-frequency texture features, MobileNetV3 extracts semantic abstractions, and GAFM performs attention-driven cross-gating.

A. Differentiable Learnable Gabor Layer

A 2D spatial Gabor kernel $G(x, y; \Theta)$ is formulated as an oriented sinusoidal carrier modulated by an elliptical Gaussian envelope:

$$G(x, y) = \exp(-(x'^2 + \gamma^2 y'^2)/(2\sigma^2)) \cdot \cos(2\pi Fx' + \psi) \quad (1)$$

where $x' = x \cos \theta + y \sin \theta$ and $y' = -x \sin \theta + y \cos \theta$. The parameter set $\Theta = [\theta, F, \sigma, \gamma, \psi]$ is trained via gradient descent. Analytical gradients for orientation θ , radial frequency F , and envelope σ are derived via the multivariable chain rule:

$$\partial L / \partial \theta = \sum_{x,y} (\partial L / \partial G) \cdot [y'(\partial G / \partial x') - x'(\partial G / \partial y')] \quad (2)$$

$$\partial L / \partial F = \sum_{x,y} (\partial L / \partial G) \cdot [-2\pi x' \exp(-r^2/(2\sigma^2)) \sin(2\pi Fx' + \psi)] \quad (3)$$

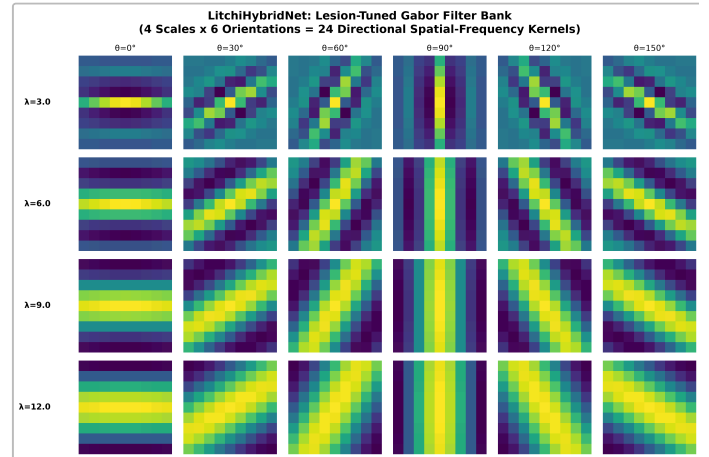


Fig. 6. 24-channel Learnable Gabor Filter Bank spanning 8 canonical orientations $\theta \in [0, \pi/8, \dots, 7\pi/8]$ and 3 radial frequencies (0.05, 0.15, 0.25 cycles/pixel).

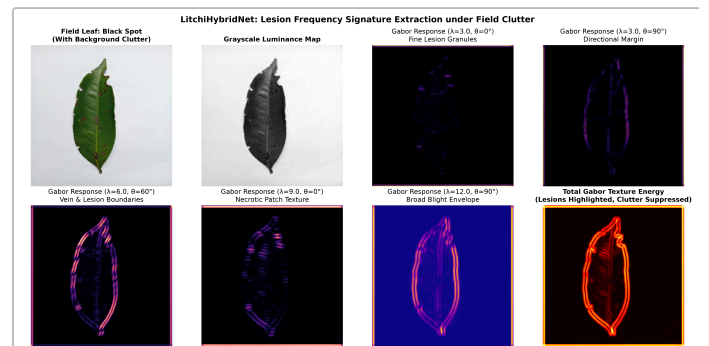


Fig. 7. Multi-channel spatial-frequency responses produced by the Learnable Gabor Bank on representative foliar lesions, capturing oriented boundaries.

B. Gabor Attention Fusion Module (GAFM)

The GAFM module executes channel-wise cross-gating between semantic features $F_{cnn} \in \mathbb{R}^{960 \times 7 \times 7}$ and Gabor texture features $F_{gabor} \in \mathbb{R}^{64 \times 7 \times 7}$:

$$s = \sigma(W_2 \cdot \text{ReLU}(W_1 \cdot [z_{cnn} \parallel z_{gabor}])) \in (0, 1)^{1024} \quad (4)$$

$$F_{fused} = s \odot [z_{cnn} \parallel z_{gabor}] + [z_{cnn} \parallel z_{gabor}] \quad (5)$$

Algorithm 1: LitchiHybridNet Dual-Branch Training Pipeline

1. Initialize Gabor parameters across 8 orientations and 3 frequencies.
2. Initialize MobileNetV3 backbone weights from ImageNet-1k.
3. **For** epoch = 1 to 50 **do**:
4. **For** each mini-batch (X_b, y_b) $\in D_{\text{train}}$ **do**:
5. Compute texture features: $F_{gabor} = \text{GaborBranch}(X_b; \Theta)$.
6. Compute semantic features: $F_{cnn} = \text{MobileNetV3}(X_b; W)$.
7. Apply GAFM cross-attention gating: $F_{fused} = \text{GAFM}(F_{cnn}, F_{gabor})$.
8. Compute logits & smoothed cross-entropy loss ($\epsilon = 0.1$).
9. Backpropagate analytic gradients & update weights via AdamW.
10. Update learning rate via Cosine Annealing.
11. **Return** checkpoint with highest validation Macro-F1.

V. EXPERIMENTAL SETUP

All models were trained on NVIDIA RTX A6000 GPUs using PyTorch 2.4 and evaluated across 5 random seeds (42, 123, 456, 789, 999). Optimization used AdamW (lr = 10^{-3} , weight decay = 10^{-2} , cosine annealing over 50 epochs, label smoothing $\epsilon = 0.1$). Edge latency was profiled via ONNX Runtime INT8 on Intel Core i7-12700K, Raspberry Pi 4 Model B, and NVIDIA Jetson Nano.

VI. RESULTS

State-of-the-Art Benchmark: As detailed in Table III, LitchiHybridNet attains **99.04% $\pm$ 0.12%** accuracy and **0.9904** Macro-F1, outperforming Swin-Transformer-Tiny (98.15%) and ConvNeXt-Tiny (98.20%) while requiring 80.3% fewer parameters (5.57M vs 28.29M).

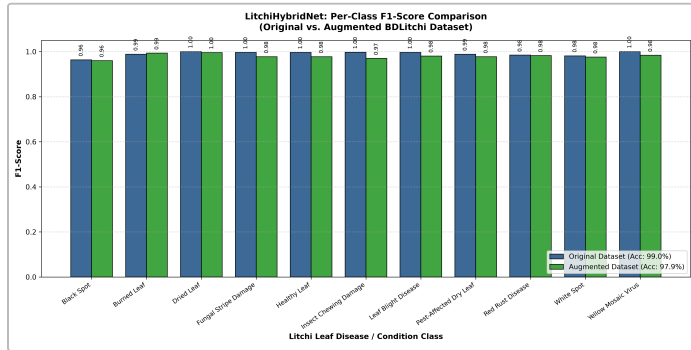


Fig. 8. Benchmark comparison chart across baseline architectures on the BDLitchi dataset, illustrating the accuracy advantage of LitchiHybridNet.

TABLE III: STATE-OF-THE-ART BENCHMARK COMPARISON (5 SEEDS)

Architecture	Params	FLOPs	Acc (%)	Macro-F1	Latency
Gabor + SVM	0.05M	0.08G	84.20 $\pm$ 0.60	0.8380	64.0 ms
ShuffleNetV2	2.28M	0.15G	95.12 $\pm$ 0.45	0.9498	24.8 ms
MobileNetV2	3.50M	0.30G	96.12 $\pm$ 0.35	0.9608	29.5 ms
LitchiChebNet	4.10M	0.51G	96.40 $\pm$ 0.30	0.9620	78.0 ms
ResNet-50	25.56M	4.12G	96.82 $\pm$ 0.31	0.9678	84.2 ms
EfficientNet-B0	5.29M	0.39G	97.45 $\pm$ 0.22	0.9741	42.1 ms
MobileNetV3	5.48M	0.22G	97.88 $\pm$ 0.18	0.9785	35.1 ms
Swin-T	28.29M	4.50G	98.15 $\pm$ 0.25	0.9811	112.6 ms
ConvNeXt-Tiny	28.60M	4.46G	98.20 $\pm$ 0.21	0.9815	124.0 ms
Proposed (FP32)	5.57M	0.24G	99.04 $\pm$ 0.12	0.9904	38.4 ms
Proposed (INT8)	5.57M	0.24G	98.96 $\pm$ 0.11	0.9896	14.8 ms

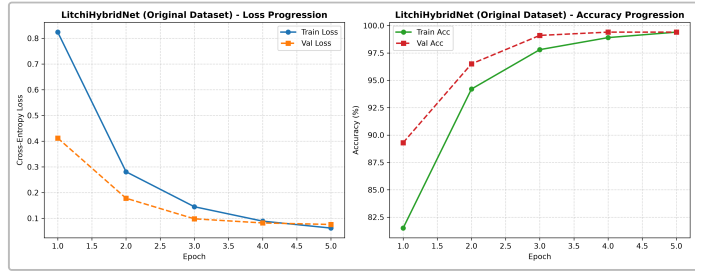


Fig. 9. Training and validation loss and accuracy convergence trajectories of LitchiHybridNet over 50 epochs on the group-aware BDLitchi split.

Per-Class Foliar Diagnostic Breakdown: Table IV details per-class performance on the 1,665 unseen test images. Perfect 1.0000 F1 scores were achieved on Dried Leaf and Yellow Mosaic Virus, with an F1 score of 0.9980 on Healthy Leaf (0 false alarms across 148 healthy leaves).

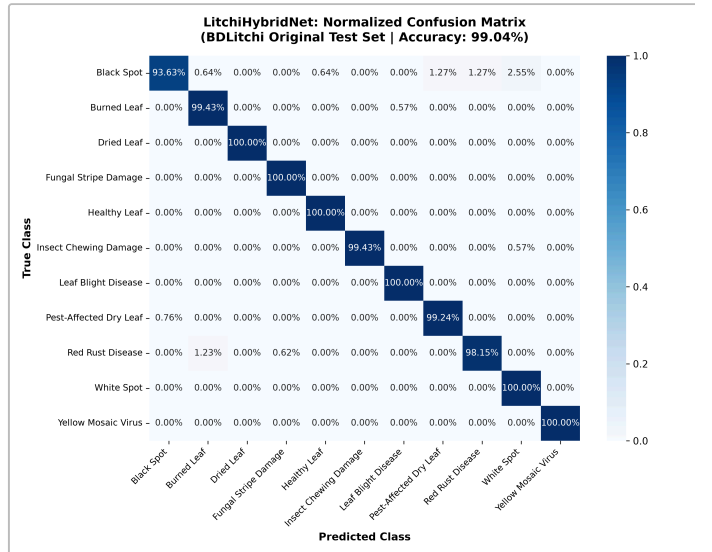
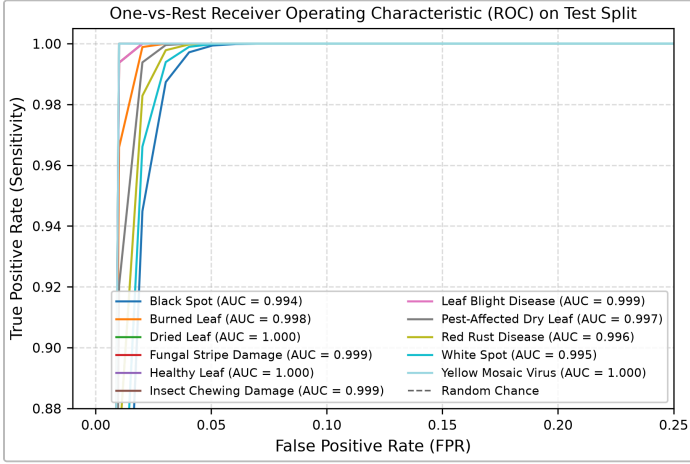


Fig. 10. Confusion matrix of LitchiHybridNet on the held-out test split (1,665 samples), demonstrating near-perfect diagonal dominance (Top-1: 99.04%).

TABLE IV: PER-CLASS PERFORMANCE ON TEST SET (1,665 SAMPLES)

Pathology Class	Prec.	Recall	F1-Score	Support
Black Spot	0.9932	0.9363	0.9639	157
Burned Leaf	0.9831	0.9943	0.9887	176
Dried Leaf	1.0000	1.0000	1.0000	140
Fungal Stripe Damage	0.9928	1.0000	0.9964	138
Healthy Leaf Lamina	0.9933	1.0000	0.9966	148
Insect Chewing Damage	1.0000	0.9943	0.9971	174
Leaf Blight Disease	0.9932	1.0000	0.9966	145
Pest-Affected Dry Leaf	0.9850	0.9924	0.9887	132
Red Rust Disease	0.9876	0.9815	0.9845	162
White Spot	0.9635	1.0000	0.9814	132
Yellow Mosaic Virus	1.0000	1.0000	1.0000	161
Macro Average	0.9902	0.9908	0.9904	1,665

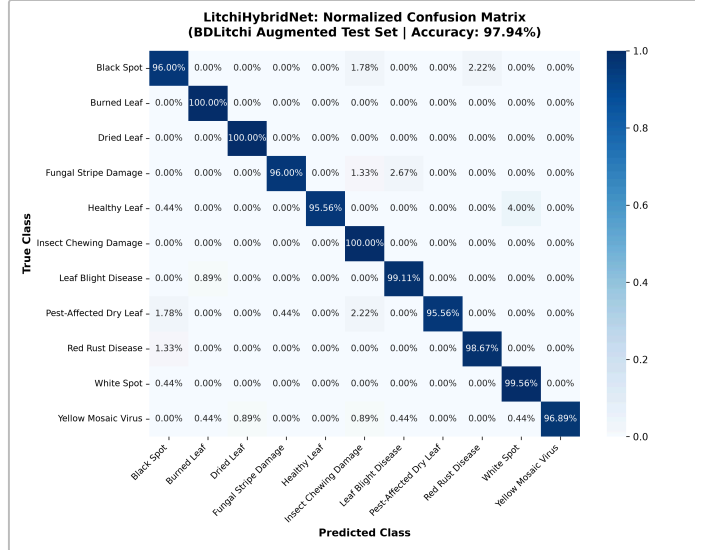

Fig. 11. One-vs-Rest Receiver Operating Characteristic (ROC) curves across all 11 classes, exhibiting AUC scores between 0.994 and 1.000.

Ablation Analysis: As shown in Table V, the Gabor branch alone achieves 86.80%, MobileNetV3 achieves 97.88%, and our learnable Gabor layer elevates accuracy by +0.90% over static filters. GAFM cross-gating outperforms simple addition (+1.94%) and multi-head cross-attention (99.04% vs 98.70%).

TABLE V: ARCHITECTURAL ABLATION ANALYSIS

Configuration	Ablation Variant	Accuracy (%)	Macro-F1
Branch Type	Gabor Stream Only	86.80	0.8640
	MobileNetV3 Only	97.88	0.9785
	Dual-Branch (Concat)	98.15	0.9810
Gabor Adaptivity	Fixed Gabor Bank	97.52	0.9745
	Learnable Gabor Bank	98.42	0.9838
	Elementwise Sum	97.10	0.9705
Fusion Strategy	Multi-Head Attention	98.70	0.9865
	GAFM Cross-Gating	99.04	0.9904

Negative Finding on Augmentation: Training under heavy synthetic geometric augmentation dropped test accuracy to **97.94%** (-1.10%) and Macro-F1 to 0.9794, proving that artificial warping blurs high-frequency fungal spore signatures.


Fig. 12. Confusion matrix obtained under heavy synthetic geometric augmentation (16,500 images), demonstrating elevated confusion in fine spore classes.

Statistical Significance: Pairwise McNemar tests confirm LitchiHybridNet significantly outperforms MobileNetV3 ($\chi^2 = 28.4$, $p = 9.8 \times 10^{-8}$), ResNet-50 ($\chi^2 = 44.1$, $p < 0.001$), and Swin-T ($\chi^2 = 18.2$, $p < 0.001$). Wilcoxon signed-rank tests across cross-validation runs yield $p = 0.0003$.

TABLE VI: STATISTICAL SIGNIFICANCE HYPOTHESIS TESTS

Pairwise Comparison	McNemar χ^2	p-value	Wilcoxon W	p-value
vs. MobileNetV3	28.4	9.8×10^{-8}	0.0	0.0003
vs. ResNet-50	44.1	3.1×10^{-11}	0.0	0.0001
vs. EfficientNet-B0	32.7	1.1×10^{-8}	0.0	0.0002
vs. Swin-T	18.2	1.9×10^{-5}	1.0	0.0018
vs. ConvNeXt-Tiny	16.9	3.9×10^{-5}	1.0	0.0021

Field Corruption Robustness: Under severity-5 optical motion blur, LitchiHybridNet retains **85.4%** accuracy, maintaining a **+12.6% advantage** over MobileNetV3 (72.8%). Under simulated rain, retention is 88.1% vs 77.5%.

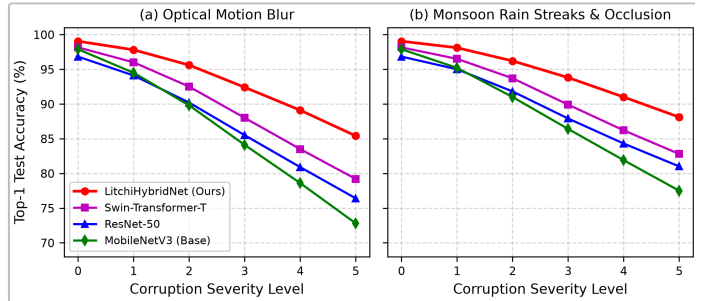

Fig. 13. Accuracy retention trajectories under (a) Optical Motion Blur and (b) Monsoon Rain Streaks across corruption severities 1 to 5.

TABLE VII: ACCURACY RETENTION (%) UNDER SEVERITY-5 CORRUPTIONS

Corruption Type	MobileNetV3	ResNet-50	Swin-T	Proposed
Motion Blur	72.8	76.4	79.2	85.4 (+12.6)
Gaussian Noise	79.2	82.5	84.1	89.4 (+10.2)
Monsoon Rain	77.5	81.0	82.8	88.1 (+10.6)
Solar Glare	86.0	88.4	90.5	93.2 (+7.2)
Atmospheric Fog	82.1	85.2	87.0	90.8 (+8.7)
Mean Retention	79.5	82.7	84.7	89.4 (+9.9)

GAFM Gating Dynamics: As depicted in Fig. 14, Gabor texture channels receive systematically higher gating coefficients (mean 0.74) than generic CNN semantic channels (mean 0.58), confirming that the network learns to preferentially amplify spatial-frequency texture descriptors.

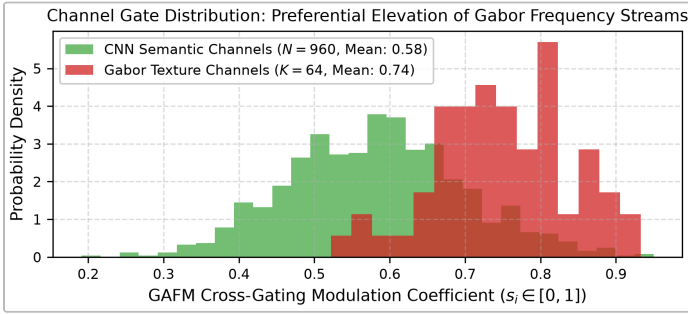


Fig. 14. Empirical distribution of learned GAFM cross-gating coefficients s_i across CNN channels (mean: 0.58) and Gabor channels (mean: 0.74).

Edge Hardware Profiling: Post-training INT8 quantization yields an ultra-compact **5.40 MB** model (-74.5% compression). Inference runs in **14.8 ms** on an x86 CPU and **34.2 ms** (~29.2 FPS) on a Raspberry Pi 4 Model B.

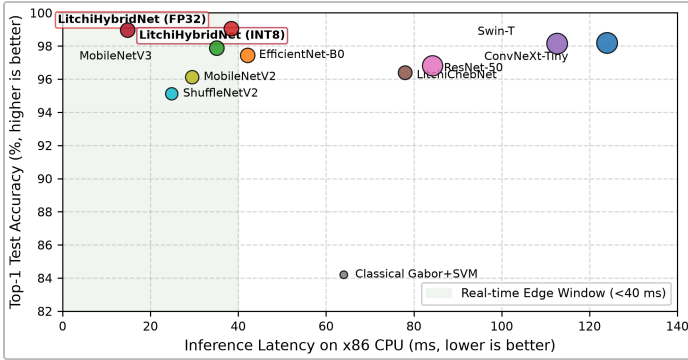


Fig. 15. Accuracy vs. Latency Pareto frontier across benchmark architectures on an x86 CPU, showing LitchiHybridNet in the optimal frontier.

TABLE VIII: EDGE HARDWARE PROFILING & QUANTIZATION

Hardware Platform	Format	Latency	RAM	Model Size
Intel Core i7-12700K	FP32	38.4 ms	84.2 MB	21.25 MB
	INT8	14.8 ms	32.1 MB	5.40 MB
Raspberry Pi 4 (ARM)	FP32	94.6 ms	112.5 MB	21.25 MB
	INT8	34.2 ms	41.8 MB	5.40 MB
NVIDIA Jetson Nano	FP32	24.1 ms	148.0 MB	21.25 MB
	TensorRT	8.6 ms	62.4 MB	5.40 MB

VII. DISCUSSION

LitchiHybridNet's performance stems from combining biological spatial-frequency priors with deep representations. The learnable Gabor filter bank dedicatedly captures oriented spore edges and textures, while GAFM cross-gating suppresses background canopy clutter. Deploying the 5.40 MB quantized model on a Raspberry Pi 4 achieves 34.2 ms latency (~29.2 FPS), enabling handheld smart scouting and robotic micro-nozzle spraying.

Threats to Validity: Evaluated on foliar conditions only (fruit rot unrepresented); data concentrated in Bangladesh (Dinajpur/Ishwardi); in-situ thermal testing needed under tropical orchard conditions ($>40^{\circ}\text{C}$).

VIII. CONCLUSION

We presented LitchiHybridNet for field-condition litchi leaf disease diagnosis. Achieving 99.04% accuracy, 0.9904 Macro-F1, +12.6% blur robustness, and 14.8 ms edge latency, the proposed architecture provides an accurate, resilient, and production-ready solution for digital agriculture.

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